2 Electrostatics–II

INTRODUCTION TO ELECTROSTATICS

- Electrostatics is a branch of physics that is essential for learning various concepts of electromagnetics and engineering.
- Electrostatics is mainly related to a static electric charge.
- An electric charge has its effect on the surrounding space or region. This surrounding space is referred to as an electric field of that charge.
- The electric field produced due to the stationary electric charge is time-invariant (does not vary with time) and referred to as the static electric field.
- **Electrostatics:** The study of the time-invariant electric field in the space or vacuum, producing various types of static charge distributions is known as electrostatics.
- An example of such a time-invariant electric field is the field utilized in a cathode ray tube (CRT) for focusing and deflecting a beam.
- Electrostatics plays a significant part in our daily life. Most PC's external devices, like keyboards, touchpads, LCDs, etc. work on the guidelines of electrostatics.
- Machines like X-ray machines and medical instruments used for electrocardiograms, scanning, and many industrial processes like spray painting, electro-deposition, etc. are also based on the principle of electrostatics.
- Electrostatics is also employed in many components such as resistors, capacitors, etc., and the device like bipolar junction transistors (BJTs), field-effect transistors (FETs) are also based on electrostatics.
- Charles Augustin de Coulomb was honoured for his research in the field of electricity and magnetism, and formulated coulomb's law. Which will be discussed later in this chapter.

Gray Matter Alert!!!



Father of Electrostatics : Charles-Augustin de Coulomb (June 1736 – 1806) was a French military engineer and physicist. He is known for the discovery of Coulomb's law. He also did work on friction. The SI unit of electric charge, coulomb (C), was named in his honor in 1880.

Rack your Brain

Name the force acting on balloons if we rub them on hair?

- The study of electrostatics is initiated by learning the two fundamental laws that governs the electrostatic fields.
  1. Coulomb's law
  2. Gauss's law
- Coulomb's law is used for determining the electric field due to any type of charge configuration.
- Gauss's law is suitable when the charge distribution is symmetrical.
- The concept of electric field intensity will be introduced and applied to the cases involving point charges, line charges, surface charges, and volume charges.
- We will assume that the electric fields are in the free space or vacuum in this chapter.
- Electrostatics is basically a theory which explains the physical phenomena related to the interaction between stationary electric charges or charge distributions in a space with stationary boundaries.

COULOMB'S LAW

- Coulomb's inverse-square law is another name of coulomb's law.
- The study of electrostatics with the experimental results was obtained by Charles Coulomb.
- The experiment was related to the force exerted between the two points-charges, which were placed near each other. This force exerted was due to the electric fields produced by the point charges.
- A charge that is located on a body whose dimensions are much shorter than other related dimensions is referred to as a point charge.
- For example, an accumulation of electric charges on a pinhead is considered as a point charge.

Rack your Brain

Try to list some examples of electrostatic forces from our daily life?

Gray Matter Alert!!!

In Coulomb's experiment, the torsion balance was an insulating rod with a metal-coated ball attached to one end, suspended by a silk thread. A charged ball (with a known charge type) was brought near the other charged ball of a similar polarity.

The two charged balls repelled each other, twisting the thread through a certain angle. Coulomb evaluated the force acting between the two balls about a given angle due to the twisted silk thread and derived his inverse-square proportionality law.

- The polarity of charge can be positive or negative.
- Charge is generally measured in coulombs (C).
  1 Coulomb = 6.241×10^{18} electrons
  and 1 electron charge (q) = -1.6021×10^{-19} C.
- If the two charges of magnitudes Q_1 and Q_2 respectively, are situated on bodies, whose dimensions are smaller compared to the separation distance R between them. Then, Coulomb's law states that the force F between the two point charges Q_1 and Q_2 will be-
  (i) Along the line joining the two point-charges Q_1 and Q_2.
  (ii) Directly proportional to the product of two charges i.e. Q_1 and Q_2.
  (iii) Inversely proportional to the square of distance R between Q_1 and Q_2.
  (iv) Attractive or repulsive, according to the like or unlike signs of the two charges.
- Mathematically, force F can be expressed as-

$$F \propto \frac{Q_1 Q_2}{R^2}$$

$$\text{or, } F = \frac{kQ_1Q_2}{R^2} \quad \ldots (2.1)$$

Where,
k = Proportionality constant
Q_1 = First point charge (in coulombs)
Q_2 = Second point charge (in coulombs)
R = Separation distance (in meters)

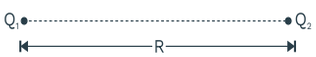


Figure 2.1

- The k (constant of proportionality) depends on the medium where the charges are located. And hence, according to the equation (2.1), the force also depends on the medium where the point charges are located.

Gray Matter Alert!!!

Electrostatic forces are non-contact forces**.** These forces pull or push the objects without touching them. Rubbing objects together results in the charge creation. The charged body either attracts or repel the other uncharged body based on the charge type without a proper contact.

Rack your Brain

Why are Coulomb's forces referred to as conservative forces?

- In equation (2.1), the point charges are in coulombs (C) and distance R is in meters(m), thus the force F is in newtons (N).
- Therefore, to satisfy Coulomb's law, the proportionality constant is defined as-

$$k = \frac{1}{4\pi\epsilon}$$

Where,
ϵ = Permittivity of the medium where point-charges are located [in Farads /meter].

Previous Year's Questions

Match the List–I (Laws) & List–II (Applications) and choose the correct answer using the codes-

List–I	**List–II**
P. Ampere's law.	(1) Force on a charge.
Q. Biot's law.	(2) Force due to a current carrying conductor.
R. Coulomb's law.	(3) Electric flux density at a point.
S. Gauss's law.	(4) Magnetic flux density at a point.

Codes:

	P	**Q**	**R**	**S**
(a)	3	2	1	4
(b)	4	2	1	3
(c)	4	1	2	3
(d)	3	1	2	4

And $\epsilon = \epsilon_0 \epsilon_r$ (in general)

Where,
ϵ= Absolute permittivity
ϵ_0 = Permittivity in free space/ vacuum
ϵ_r = Relative permittivity

- Also, for the free space $\epsilon = \epsilon_0$

and, $\epsilon_r = 1$

Thus, $\epsilon = \epsilon_0$

So, equation (2.1) can be rewritten as –

Force, $F = \frac{1}{4\pi \epsilon_0} \frac{Q_1 Q_2}{R^2}$ ... (2.2)

- Hence, by Coulomb's law-

$$F = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2} \quad \text{... (2.3)}$$

where, $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} = 8.854 \times 10^{-12}$ F/m

$$k = \frac{1}{4\pi \epsilon_0} = 8.898 \times 10^{-9} \simeq 9 \times 10^9 \text{m/F}$$

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The Coulomb constant(k) (or the electric force constant, or the electrostatic constant) is a proportionality constant in electrostatics equations. It is equal to 8.9875 × 10^9 $kgm^3s^{-2}C^{-2}$

COULOMB'S LAW IN VECTOR FORM

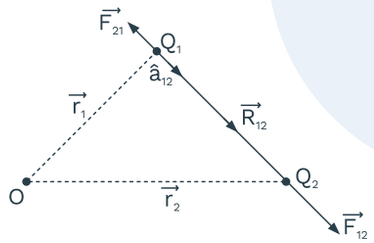


Figure 2.2: Vector form of Coulomb's law

- The force $\vec{F}$ between the two point-charges has a fixed direction that is a straight line joining two charges.
- In figure 2.2, the force exerted between charges Q_1 and Q_2 is expressed in vector form.
- Considering Q_1 and Q_2, the two point-charges are located at the points with two position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively.
- Thus, the force applied by Q_1 on Q_2 will act along the direction $\vec{R}_{12}$ where $\hat{a}_{12}$ is a unit vector along $\vec{R}_{12}$.
- The Coulomb's law in vector form will be expressed as-

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2} \hat{a}_{12} \quad \text{....(2.4)}$$

Rack your Brain

Can coulomb's force be infinite?

$\vec{F}_{12}$ = Force by Q_1 on Q_2

Where,

$$\hat{a}_{12} = \frac{\vec{R}_{12}}{\left|\vec{R}_{12}\right|} = \frac{\text{Vector}}{\text{Magnitude of vector}} = \frac{\vec{R}}{R} \quad ...(2.5)$$

$$\hat{a}_{12} = \frac{\vec{R}_{12}}{R} = \frac{\vec{r}_2 - \vec{r}_1}{\left|\vec{R}_{12}\right|} = \frac{\vec{r}_2 - \vec{r}_1}{\left|\vec{r}_2 - \vec{r}_1\right|} \quad ...(2.6)$$

As, $R = \left|\vec{R}_{12}\right|$ and $\vec{R}_{12} = |\vec{r}_2 - \vec{r}_1|$.

where $\left|\vec{R}_{12}\right|$ = magnitude of $\vec{R}_{12}$ (Distance between the two point-charge Q_1 and Q_2).

- By substituting, equations (2.5) and (2.6) in equation (2.4), we get-

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi \in_0} \frac{\vec{R}_{12}}{R^3}$$

$$\vec{F}_{12} = \frac{Q_1 Q_2 \left(\vec{r}_2 - \vec{r}_1\right)}{4\pi \in_0 \left|\vec{r}_2 - \vec{r}_1\right|^3} \quad ... (2.7)$$

- As shown in figure 2.2, the force $\vec{F}_{21}$ by Q_2 on Q_1 will be expressed as-

$$\vec{F}_{21} = \left|\vec{F}_{21}\right| \hat{a}_{21} \quad ... (2.8)$$

And $$\hat{a}_{21} = \frac{\vec{r}_1 - \vec{r}_2}{\left|\vec{r}_1 - \vec{r}_2\right|} = \frac{\vec{R}_{21}}{\left|\vec{R}_{21}\right|} = \frac{\vec{R}_{21}}{R}$$

Also, $$\vec{r}_1 - \vec{r}_2 = -\left[\vec{r}_2 - \vec{r}_1\right]$$

$$\hat{a}_{21} = -\left[\hat{a}_{12}\right] \quad ... (2.9)$$

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The coulomb law is generally applicable only to the point charges and does not apply to the two charged bodies of finite sizes like two charged spheres.

Rack your Brain

If the force between two charges is 100N. And if the distance between them is tripled, then what will be the new force?

Previous Year's Questions

Consider the two equal and negative charges (-q) that are fixed at points (0, -a) and (0, a) on y-axis respectively. And a positive charge +Q was released from rest at the point (2a, 0) on the x-axis. Then the charge Q will-
(i) Execute simple harmonic motion about the origin.
(ii) Move to the origin remain at rest.
(iii) Move to infinity.
(iv) Execute the oscillatory but not simple harmonic motion.

- Now, substituting equation (2.9) in equation (2.8)-

$$\vec{F}_{21} = \frac{Q_1 Q_2}{4\pi \in_0 R^2} \hat{a}_{21} = \frac{Q_1 Q_2}{4\pi \in_0 R^2} \left(-\hat{a}_{12}\right)$$

$$\vec{F}_{21} = -\frac{Q_1 Q_2}{4\pi \in_0 R^2} \hat{a}_{12} = -\vec{F}_{12} \quad \ldots (2.10)$$

Or $\vec{F}_{21} = -\left(\vec{F}_{12}\right)$ ... (2.11)

- From equation (2.11), it is concluded that the force exerted by two charges Q_1 and Q_2 on each other is equal in magnitude but opposite in direction.



Figure 2.3: (a) and (b) shows that like charges will repel. (c) Unlike charges will attract each other.

- For more than two points charges, the principle of superposition is employed to determine the force acting on a specific charge.
- Considering N charges Q_1, Q_2 ..... Q_N which are located respectively at points, with position vectors $\vec{r}_1, \vec{r}_2 \ldots\ldots\ldots \vec{r}_N$.

  Thus, the resultant force $\vec{F}$ on a charge Q at a point of distance r will be-

Rack your Brain

The value of the electric field (V/m) at 1 m distance from an infinite line with charge density 1 C/m is-
(A) $2\pi \in_0$
(B) $(1/2\pi\in_0)$
(C) $(\in_0/2\pi)$
(D) $(2\pi/\in_0)$

$$\vec{F} = \frac{QQ_1}{4\pi \in_0}\frac{(\vec{r}-\vec{r}_1)}{|\vec{r}-\vec{r}_1|^3} + \frac{QQ_2}{4\pi \in_0}\frac{(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_2|^3} + ... + \frac{QQ_N}{4\pi \in_0}\frac{(\vec{r}-\vec{r}_N)}{|\vec{r}-\vec{r}_N|^3} \quad ...(2.12)$$

$$\vec{F} = \frac{Q}{4\pi \in_0}\sum_{k=1}^{N}\frac{Q_k(\vec{r}-\vec{r}_k)}{|\vec{r}-\vec{r}_k|^3} \quad ...(2.13)$$

where k is an integer and $1 \le k \le N$.

Limitation of Coulomb's law:

- It's difficult to apply when charges are of arbitrary shape (except point charge).

Example 1:

Consider a charge Q_1 = 0.03 mC located at M(1, 2, 2) and another charge $Q_2 = -10^{-3}$ C is located at N(2, 0, 4) in a vacuum. The magnitude of the force exerted on Q_2 due to Q_1 will be-

(A) 30 N (B) 40 N
(C) 10 N (D) –10 N

Solution:

Given, Two-point charges-

$Q_1 = 0.03 \text{ mC} = 0.03 \times 10^{-3}$ C

$Q_2 = -10^{-3}$ C

The perpendicular distance, $\vec{R}_{12}$ will be-

$$\vec{R}_{12} = \vec{r}_2 - \vec{r}_1$$

$$\vec{R}_{12} = (2-1)\hat{a}_x + (0-2)\hat{a}_y + (4-2)\hat{a}_z$$

$$\vec{R}_{12} = \hat{a}_x - 2\hat{a}_y + 2\hat{a}_z$$

and magnitude $|\vec{R}_{12}| = \sqrt{(1)^2 + (-2)^2 + (2)^2}$

$$|\vec{R}_{12}| = \sqrt{1+4+4}$$

$$|\vec{R}_{12}| = \sqrt{9} = 3$$

and $\hat{a}_{12} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{1}{3}(\hat{a}_x - 2\hat{a}_y + 2\hat{a}_z)$

The force exerted on Q_2 charge due to Q_1 will be expressed as –

Previous Year's Questions

The electric field intensity of the conductor is normal to the surface it, and is also-

a) equal to $\in$ times the surface charge density at that point,
b) equal to $(1/\in)$ times the surface charge density at that point,
c) equal to surface charge density at that point,
d) cannot be calculated.

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi \in_0 R_{12}^2} \hat{a}_{12}$$

$$\vec{F}_{12} = \frac{\left(0.03 \times 10^{-3}\right)\left(-10^{-3}\right)}{4\pi\left(\frac{10^{-9}}{36\pi}\right)(3)^2} \times \frac{1}{3}\left(\hat{a}_x - 2\hat{a}_y + 2\hat{a}_z\right)$$

$$\vec{F}_{12} = \frac{-0.03 \times 36\pi \times 10^{-6}}{4\pi \times 9 \times 10^{-9}} \times \frac{\left(\hat{a}_x - 2\hat{a}_y + 2\hat{a}_z\right)}{3}$$

$$\vec{F}_{12} = -30\left(\frac{\hat{a}_x - 2\hat{a}_y + 2\hat{a}_z}{3}\right) N$$

The magnitude of the force exerted on Q_2 due to Q_1 will be 30 N and the vector $\vec{F}_{12}$ will be described as–

$$\vec{F}_{12} = -10\hat{a}_x + 20\hat{a}_y - 20\hat{a}_z \text{ N}$$

$$|\vec{F}_{12}| = \sqrt{(10)^2 + (20)^2 + (20)^2}$$

$$= 30$$

Hence, the correct option is (A).

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Which of the following is the condition for the validity of Coulomb's law?

a. The charges must have a spherically symmetric distribution.
b. Charges must be distinct point-charges.
c. Charges must be stationary w.r.t each other.

ELECTRIC FIELD INTENSITY

- The force experience by a unit positive charge when placed in an electric field is called the electric field intensity $\vec{E}$ (or electric field strength).
- Then, the electric field intensity $\vec{E}$ due to point-charge Q is defined as the force per unit charge on Q, when placed in an electric field.

$$\vec{E} = \frac{\vec{F}}{Q} \quad \text{..... (2.14)}$$

Or $\vec{E} = \lim_{Q \to 0}\left(\frac{\vec{F}}{Q}\right)$

- The force experienced by Q_2 due to the field of Q_1 is given by-

Definitions

A system in which the infinite numbers of charges are closely packed and are uniformly distributed over the conductor is referred to as the continuous charge distribution system.

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 |R_{12}|^2} \hat{a}_{R_{12}} \quad \text{..... (2.15)}$$

- Let $Q_1 = Q$ such that $\vec{F}_{12} = \vec{F}, \hat{a}_{R_{12}} = \hat{a}_R$ and, $|\vec{R}_{12}| = R$
- Thus, the force experienced by Q_2 due to the field of Q will be written as-

$$\vec{F} = \frac{Q Q_2}{4\pi \epsilon_0 R^2} \hat{a}_R \quad \text{..... (2.16)}$$

- And the electric field intensity $\vec{E}$, will be-

$$\vec{E} = \frac{\vec{F}}{Q_2} = \frac{Q Q_2}{4\pi \epsilon_0 R^2} \left(\frac{1}{Q_2}\right) \hat{a}_R$$

$$\vec{E} = \frac{Q}{4\pi \epsilon_0 R^2} \hat{a}_R \text{ (V/m)} \quad \text{..... (2.17)}$$

Where,

$\vec{E}$ = Electric field intensity (N/C) or (V/m)

Q = Point charge (in C)

R = Magnitude of vector of $\vec{R}$ and

$\hat{a}_R$ = Unit vector (in the $\vec{R}$ direction)

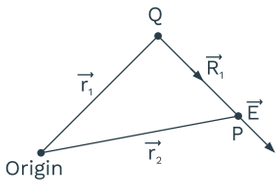


Figure 2.4: Electric field intensity at P due to charge Q

- For N point-charges Q_1, Q_2, ... Q_N located at $\vec{r}_1, \vec{r}_2 \ldots\ldots \vec{r}_N$, the electric field intensity $\vec{E}$ at point $\vec{r}$ is described as follows-
From equations (2.12) and (2.14),

$$\vec{E} = \frac{Q_1(\vec{r} - \vec{r}_1)}{4\pi \epsilon_0 |\vec{r} - \vec{r}_1|^3} + \frac{Q_2(\vec{r} - \vec{r}_2)}{4\pi \epsilon_0 |\vec{r} - \vec{r}_2|^3} + \ldots\ldots + \frac{Q_N(\vec{r} - \vec{r}_N)}{4\pi \epsilon_0 |\vec{r} - \vec{r}_N|^3}$$

$$\vec{E} = \frac{1}{4\pi \epsilon_0} \sum_{k=1}^{N} \frac{Q_k(\vec{r} - \vec{r}_k)}{|\vec{r} - \vec{r}_k|^3} \quad \text{.....(2.18)}$$

Gray Matter Alert!!!

The electric field for the line charge is obtained by summing the electric fields for all the infinitely small point charges.

Salient Features of Electric Field Intensity:

1. It is a vector quantity.
2. It is in the direction of the force.
3. Magnitude of $\vec{E}$ depends on force's magnitude and the charge on which force is acting.
4. It depends on the permittivity ($\in$) of the medium and the distance of charges from each other.
5. It depends on the location of charges.
6. It always starts from a positive charge and ends at a negative charge.

Example 2:

The electric field $(\vec{E})$ V/m at the point (1, 1, 0) due to a point charge of 1.5 μC located at (–1, 1, 1) is-

(A) $1206\left(2\hat{a}_x - \hat{a}_z\right)$ (B) $1206\left(2\hat{a}_x + \hat{a}_z\right)$

(C) $603\left(\hat{a}_x - 2\hat{a}_z\right)$ (D) $-603\left(\hat{a}_x + 2\hat{a}_z\right)$

Solution:

Given that, a point charge, Q = 1.5 μC is located at (–1, 1, 1).

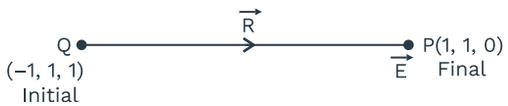
Thus, the electric field will be–

$$\vec{E} = \frac{kQ}{R^2}\hat{a}_R = \frac{1}{4\pi \in_0}\frac{Q}{R^2}\left(\frac{\vec{R}}{R}\right)$$

$$\vec{E} = \frac{1}{4\pi \in_0}\frac{\left(1.5\times 10^{-6}\right)\vec{R}}{R^3}$$

and $\vec{R} = 2\hat{a}_x - \hat{a}_z$

$$R = \left|\vec{R}\right| = \sqrt{5}$$



Therefore, $\vec{E} = \frac{1}{4\pi \in_0}\frac{\left(1.5\times 10^{-6}\right)\left(2\hat{a}_x - \hat{a}_z\right)}{\left(\sqrt{5}\right)^3}$

$$\vec{E} = \frac{1.5\times 10^{-6}}{1243.39\times 10^{-12}}\left(2\hat{a}_x - \hat{a}_z\right)$$

Gray Matter Alert!!!

The electric field is a vector quantity that has both magnitude and direction whereas the electric field intensity is the magnitude of the electric field vector.

$$\vec{E} \simeq 1206\left(2\hat{a}_x - \hat{a}_z\right)$$

Hence, the correct option is (A).

Electric fields due to various continuous charge distributions

- In most of the electromagnetic applications, the group of charged particles like charged plates, wires, or spheres, etc., can be illustrated as line, surface, or volume charge distributions.
- The electric fields $\left(\vec{E}\right)$ of a charge distribution can be analyzed by observing the distribution, as a group of differential charge elements (point charges).

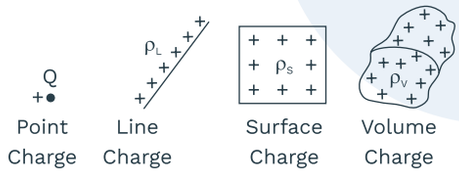


Figure 2.5: Different charge distributions.

- The continuous charge distribution along a line, on a surface or in a volume are shown in figure 2.5.
- The line charge density is denoted by ρ_L (in C/m) and the differential charge element (dQ) is given by-

  $dQ = \rho_L\, dl$

  and total charge Q will be-

  $$Q = \int_L \rho_L dl \qquad \text{(Line charge)} \qquad \ldots.(2.19)$$

- The electric field intensity $(\vec{E})$ due to ρ_L charge distribution is obtained by replacing Q in equation (2.17) with charge element $dQ = \rho_L dl$ and integrating it as follows-

  $$\vec{E} = \int_L \frac{\rho_L dl}{4\pi \epsilon_0 R^2} \hat{a}_R \quad \text{(Line charge)} \qquad \ldots..(2.20)$$

Rack your Brain

What will be the electric field due to the semicircular ring?

- Similarly, the surface charge density is denoted by ρ_S (in C/m^2). And the differential charge element (dQ) will be-

  $dQ = \rho_S \, dS$

  and total charge Q is given by-

  $$Q = \int_S \rho_S dS \qquad \text{(Surface charge)} \qquad .....(2.21)$$

- The electric field intensity $(\vec{E})$ due to ρ_S surface charge will be given by-

  $$\vec{E} = \int_S \frac{\rho_S dS}{4\pi \in_0 R^2} \hat{a}_R \quad \text{(Surface charge)} \; .....(2.22)$$

- And in case of volume charge density ρ_V (in C/m^3), the differential charge element (dQ) is given by-

  $$dQ = \rho_V \, dV$$

  And total charge Q will be-

  $$Q = \int_V \rho_V dV \qquad \text{(Volume charge)} \qquad .....(2.23)$$

- The electric field intensity due to volume charge distribution is given by-

  $$\vec{E} = \int_V \frac{\rho_V dV}{4\pi \in_0 R^2} \hat{a}_R \quad \text{(Volume charge)} \; ...(2.24)$$

  Note: The values of R and $\hat{a}_R$ may vary, as different integrals are evaluated in equation (2.20), equation (2.22) and equation (2.24).

Electric Field due to an infinite line charge

- Consider an infinitely long straight line carrying uniform line charge density (ρ_L) and placed along the z-axis.
- Assume point A is on the y-axis at which $\vec{E}$ is to be evaluated and r is the distance between point A and origin.

Rack your Brain

Choose the correct statement -

1. If the point has a charge, then the electric field is discontinuous at the point.
2. The continuous electric field at a point is zero.
3. The continuous electric field at a point if a charge is present at the point.
4. The electric field is discontinuous if a negative charge is present at the point.

Previous Year's Questions

If a uniform line charge with linear density (λ) lies along the y-axis. What flux crosses a spherical surface centered at the origin with r = R

a. $2R\lambda/\epsilon_0$
b. $R\lambda/\epsilon_0$
c. λ/ϵ_0
d. none of the above

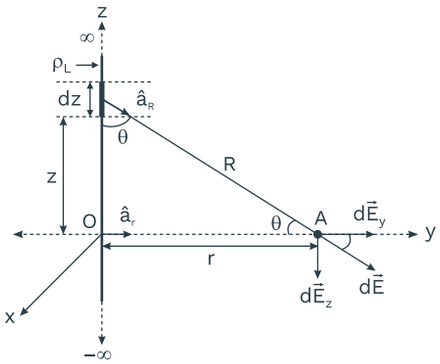


Figure 2.6: Electric field due to infinite line charge.

- Dividing z-axis into small segments dz (or dl), then charge on dz will be given by-

  $dQ = \rho_L\, dl = \rho_L\, dz$

- And the electric field element $(d\vec{E})$ will be expressed as-

$$d\vec{E} = \frac{dQ}{4\pi \in_0 (r^2+z^2)}\hat{a}_R = \frac{\rho_L dz}{4\pi \in_0 (r^2+z^2)}\hat{a}_R$$

- From the symmetry, the total electric field will have only r-component along the y-axis $(\hat{a}_r)$ or $(\hat{a}_y)$ and is given by-

$$\therefore \quad d\vec{E}_r = \frac{\rho_L dz}{4\pi \in_0}\cdot\frac{r}{(r^2+z^2)^{3/2}}\hat{a}_r$$

where $\hat{a}_R = \dfrac{r}{\sqrt{r^2+z^2}}\hat{a}_r$

and therefore $\vec{E}$ is expressed as-

$$\vec{E} = \frac{\rho_L r}{4\pi \in_0}\int_{-\infty}^{\infty}\frac{dz}{(r^2+z^2)^{3/2}}\hat{a}_r \qquad \text{..... (2.25)}$$

- Let, $z = r\tan\theta$, then $dz = r\sec^2\theta\, d\theta$.

$$\vec{E}_r = \frac{\rho_L r}{4\pi \in_0}\int_{-\pi/2}^{\pi/2}\frac{r\sec^2\theta d\theta}{[r^2(1+\tan^2\theta)]^{3/2}}\hat{a}_r$$

$$\vec{E}_r = \frac{\rho_L}{4\pi \in_0 r}\int_{-\pi/2}^{\pi/2}\cos\theta d\theta\ \hat{a}_r = \frac{\rho_L}{2\pi \in_0 r}\hat{a}_r\,(V/m)$$

Previous Year's Questions

If an interface (z = 0) between two general materials contains a uniform surface charge density $\rho_s = 10^{-9}$ C/m^2. Electric field intensity in material-A covering the left part of an interface (z < 0) with $\in_1 = 4\in_0$, is-
$\vec{E} = 10\hat{a}_x + 2\hat{a}_y - 10\hat{a}_z$ V / m.

The electric field intensity (V/m) in material-B occupying the right of interface (z > 0) with $\in_2 = 2\in_0$ will be-

(A) $10\hat{a}_x + 2\hat{a}_y - 20\hat{a}_z$
(B) $5\hat{a}_x + \hat{a}_y - 10\hat{a}_z$
(C) $10\hat{a}_x + 2\hat{a}_y - 76.5\hat{a}_z$
(D) $10\hat{a}_x + 2\hat{a}_y - 36.5\hat{a}_z$

Thus, the electric field intensity due to infinite line charge is expressed as-

$$\vec{E} = \frac{\rho_L}{2\pi \epsilon_0 r} \hat{a}_r \text{ (V/m)} \quad \text{.....(2.26)}$$

Where,

r = Perpendicular distance of point A from the line charge

$\hat{a}_r$ = Unit vector in the direction of r

ρ_L = Line charge density (in C/m)

Electric field due to circular ring

- Considering a circular ring with the radius r and uniform line charge density ρ_L (C/m). The point A is on the z-axis.
- Dividing the ring into small angular segments $d\phi$.
- The length of such segment will be-

$$dl = r\, d\phi$$

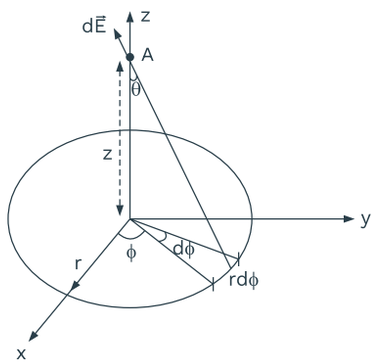


Figure 2.7: Electric field due to circular ring.

- The charge on the segment is $dQ = \rho_L\, dl = \rho_L\, r\, d\phi$.
- The electric field $\vec{E}$ due to the segment has magnitude expressed as-

$$d\vec{E} = \frac{dQ}{4\pi \epsilon_0 (r^2 + z^2)} \hat{a}_R = \frac{\rho_L r d\phi}{4\pi \epsilon_0 (r^2 + z^2)} \hat{a}_R$$

where, $|\vec{R}| = \sqrt{r^2 + z^2}$

Rack your Brain

Two large metal sheets having surface charge density $+\sigma$ and $-\sigma$ are kept parallel to each other at a small separation distance d. The electric field at any point in the region between the plates is-

(a) σ/ϵ_0 (b) $\sigma/2\epsilon_0$

(c) $2\sigma/\epsilon_0$ (d) $\sigma/4\epsilon_0$

$$\hat{a}_R = \frac{\vec{R}}{|\vec{R}|} = \frac{-r\hat{a}_r + z\hat{a}_z}{\sqrt{r^2 + z^2}}$$

$$d\vec{E} = \frac{dQ}{4\pi \in_0 (r^2 + z^2)} \cdot \frac{-r\hat{a}_r + z\hat{a}_z}{\sqrt{r^2 + z^2}}$$

$$d\vec{E} = \frac{\rho_L r d\phi}{4\pi \in_0 (r^2 + z^2)^{3/2}} \left[-r\hat{a}_r + z\hat{a}_z\right]$$

- Neglecting r-component as it will cancel each other.
  The total electric field is expressed as-

$$\vec{E}_z = \int_0^{2\pi} \frac{\rho_L r d\phi}{4\pi \in_0 (z^2 + r^2)^{3/2}} z\hat{a}_z$$

$$\vec{E}_z = \frac{\rho_L r z}{4\pi \in_0 (r^2 + z^2)^{3/2}} (2\pi - 0)\hat{a}_z = \frac{\rho_L r z}{2 \in_0 (r^2 + z^2)^{3/2}} \hat{a}_z$$

$$\vec{E} = \frac{\rho_L r z}{2 \in_0 (r^2 + z^2)^{3/2}} \hat{a}_z \quad \text{V/m} \qquad \ldots.(2.27)$$

where r = radius of the ring.
z = perpendicular distance of point A from the ring along the axis of the ring.

Rack your Brain

What are three properties of electric field lines?

Electric field intensity due to infinite surface charge distribution

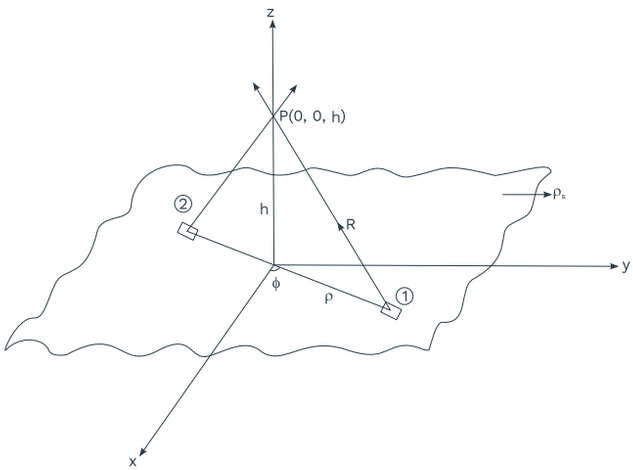


Figure 2.8: Electric field due to infinite sheet of charge

- Assume an infinite sheet of charge with uniform charge density (ρ_S) in C/m^2, placed in the x-y plane.
- The point A at which $\vec{E}$ is to be evaluated by using cylindrical coordinates is on z-axis.
- Consider the differential surface area dS carrying a charge dQ.
  or dQ = ρ_S dS
- The normal direction to dS is z-axis thus, dS is normal to z-direction and equal to dS= ρ dρ dϕ.
- Therefore, dQ = ρ_S dS = $\rho_S \rho$ dρ dϕ.
- The electric field is given by-

$$d\vec{E} = \frac{dQ}{4\pi \epsilon_0 R^2}\hat{a}_R = \frac{\rho_S \rho\, d\rho\, d\phi}{4\pi \epsilon_0 R^2}\hat{a}_R$$

where $\vec{R} = \rho(-\hat{a}_\rho) + h\hat{a}_z$

$$R = |\vec{R}| = [\rho^2 + h^2]^{1/2}$$

$$\hat{a}_R = \frac{\vec{R}}{|\vec{R}|} = \frac{-\rho\hat{a}_\rho + h\hat{a}_z}{\sqrt{\rho^2 + h^2}}$$

$$d\vec{E} = \frac{\rho_S \rho\, d\rho\, d\phi}{4\pi \epsilon_0} \cdot \left[\frac{-\rho\hat{a}_\rho + h\hat{a}_z}{\left(\rho^2 + h^2\right)^{3/2}}\right] \quad \text{.... (2.28)}$$

- There is a radial symmetry about the z-axis and all $\hat{a}_\rho$ components cancel each other.

  Thus, $\vec{E}$ will not have any radial component.
- The total electric field is expressed by integrating the above equation (2.28), we get-

$$\vec{E} = \int_s d\vec{E}_z = \frac{\rho_S}{4\pi \epsilon_0}\int_{\phi=0}^{2\pi}\int_{\rho=0}^{\infty}\frac{h\rho\, d\rho\, d\phi}{(\rho^2 + h^2)^{3/2}}\hat{a}_z \quad \text{.... (2.29)}$$

$$\vec{E} = \frac{\rho_S h}{4\pi \epsilon_0} 2\pi \int_0^{\infty}\left[\rho^2 + h^2\right]^{-3/2} \cdot \frac{1}{2} d(\rho^2)\,\hat{a}_z$$

$$\vec{E} = \frac{\rho_S h}{2 \epsilon_0}\left\{-\left[\rho^2 + h^2\right]^{-1/2}\right\}_0^{\infty}\hat{a}_z$$

$$\vec{E} = \frac{\rho_S}{2 \epsilon_0}\hat{a}_z \quad \text{.... (2.30)}$$

Gray Matter Alert!!!

SI units of electric flux is volt meters (V m), or (N m^2 C^{-1}). And the SI base units are kg·m^3·s^{-3}·A^{-1}. So, dimensional formula is [L^{3}MT^{-3}I^{-1}].

Previous Year's Questions

An induced "emf around a closed path 'C' is equal to the negative of time rate of change of the magnetic flux enclosed by that path", relates-

(A) Gauss' law (B) Ampere's law
(C) Ohm's law (D) Faraday's law

- The generalized form if $\hat{a}_n$ is in normal direction the surface containing charge, the electric field equation (2.30) can be rewritten as-

$$\vec{E} = \frac{\rho_S}{2 \epsilon_0} \hat{a}_n \text{ V/m} \qquad \text{....(2.31)}$$

Example 3:

A uniform sheet charge with $\rho_S = \frac{1}{3\pi} \text{nC/m}^2$ is located at z = 5 plane and the uniform line charge $\rho_L = \frac{-36}{5} \text{nC/m}$ is located at z = –3 and y = 3.

The electric field $\vec{E}$ at point T (x, –1, 0) will be–

(A) $18.13\,\hat{a}_x - 14.13\,\hat{a}_y$

(B) $20.73\,\hat{a}_y - 21.55\,\hat{a}_z$

(C) $14.51\hat{a}_z - 1.3\,\hat{a}_y$

(D) $-20.73\,\hat{a}_y + 21.55\,\hat{a}_z$

Solution:

Given, Uniform sheet charge $\rho_S = \frac{1}{3\pi} \text{nC/m}^2$ is located at z = 5.

Uniform line charge, $\rho_L = \frac{-36}{5} \text{nC/m}$ is located at z = –3 and y = 3.

The electric field due to surface charge is given by –

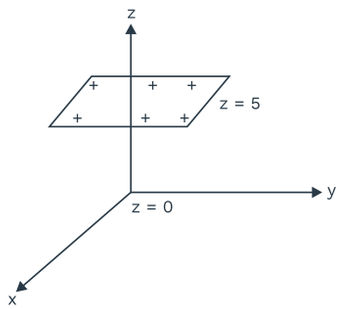
$$\vec{E}_1 = \frac{\rho_S}{2 \epsilon_0} \hat{a}_n$$

$$\vec{E}_1 = \frac{10^{-9}}{3\pi \times 2 \times \left(\frac{10^{-9}}{36\pi}\right)}$$

$$\therefore \vec{E}_1 = 6\left(-\hat{a}_z\right)$$

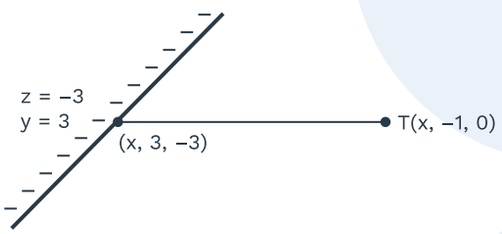
Gray Matter Alert!!!

Joseph-Louis Lagrange formulated the Gauss's law in 1773. It was then supported by Carl Friedrich Gauss in 1835. One of Maxwell's four equations, it laid the foundation for classical electrodynamics.



Now, the electric field at T(x, –1, 0) due to line charge –

$$\therefore \quad \vec{E}_2 = \frac{\rho_L}{2\pi \in_0 r}\hat{a}_r = \frac{\rho_L \vec{r}}{2\pi \in_0 r^2}$$



$$\therefore \quad \vec{E}_2 = \frac{\rho_L}{2\pi \in_0 r}\hat{a}_r$$

$$\vec{E}_2 = \frac{-36 \times 10^{-9}}{5 \times 2\pi \times \frac{10^{-9}}{36\pi}} \frac{\left(-4\hat{a}_y + 3\hat{a}_z\right)}{25}$$

$$\vec{E}_2 = -5.184\left(-4\hat{a}_y + 3\hat{a}_z\right)$$

$$= 20.736\,\hat{a}_y - 15.552\,\hat{a}_z$$

∴ The net electric field at point T(x, –1, 0) will be–

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$\vec{E} = \left(-6\hat{a}_z\right) + \left(20.736\hat{a}_y - 15.552\hat{a}_z\right)$$

$$\vec{E} = 20.736\hat{a}_y - 21.552\hat{a}_z$$

Hence, the correct option is (B).

Gray Matter Alert!!!

Gauss's law can be used to derive Coulomb's law and vice versa.

Previous Year's Questions

Two charges of magnitudes –2Q and + Q are located at points (a, 0) and (4a, 0), respectively. What is the electric flux due to these charges through a sphere of radius 4a with its center at the origin?

Electric field due to volume charge distribution

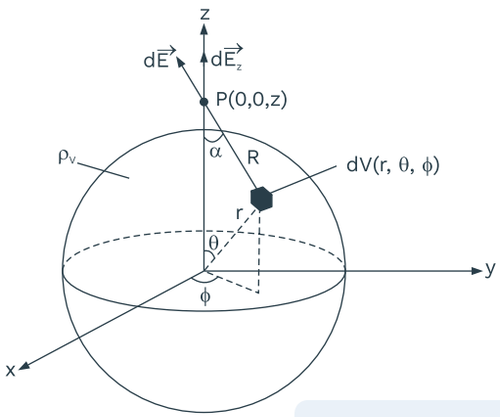


Figure 2.9

- Consider a sphere of radius 'a' centered at the origin. Let the volume of the sphere be filled uniformly with ρ_V (Volume charge density (in C/m^3)).
- The charge dQ associated with element volume dV chosen at (r, θ, ϕ)

  $dQ = \rho_V \, dV$

  and total charge, $Q = \rho_V \frac{4\pi a^3}{3}$

  where, a = radius of the sphere
- The electric field $d\vec{E}$ outside sphere at P(0, 0, z) due to elementary volume charge is-

  $$d\vec{E} = \frac{\rho_V \, dV}{4\pi \in_0 R^2} \hat{a}_R$$

  where, $\hat{a}_R = \cos\alpha \, \hat{a}_z + \sin\alpha \, \hat{a}_\rho$.
- Due to symmetry of charge distribution, the E_x and E_y will be added, then E_z will be present only.

  $$E_z = \vec{E} \cdot \hat{a}_z = \int_V dE \cos\alpha = \frac{\rho_V}{4\pi \in_0} \int_V \frac{dV \cos\alpha}{R^2}$$

- Applying the cosine rule-
  $R^2 = z^2 + r^2 - 2z \, r \cos\theta$
  $r^2 = z^2 + R^2 - 2z \, R \cos\alpha$

Rack your Brain

The Gaussian surface for a line charge will be-
(i) sphere (ii) cylinder
(iii) cube (iv) cuboid

Previous Year's Questions

A charge q is placed at the center of a cube of side L. What is the electric flux passing through each face of the cube?

and $dV = r^2 \sin\theta \, dr \, d\theta \, d\phi$

and $\cos\alpha = \dfrac{z^2 + R^2 - r^2}{2zR}$

$\cos\theta = \dfrac{z^2 + r^2 - R^2}{2zr}$

Differentiating above equation-

$\sin\theta \, d\theta = \dfrac{RdR}{zr}$

- As $0 < \theta < \pi$, R varies from (z – r) to (z + r), if point P is outside the sphere.

Thus, electric field $(\vec{E})$ is given by-

$$E_z = \frac{\rho_V}{4\pi \in_0} \int_{\phi=0}^{2\pi} d\phi \int_{r=0}^{a} \int_{R=z-r}^{z+r} \frac{r^2 RdR}{zr} dr \left[\frac{z^2 + R^2 - r^2}{2zR}\right] \frac{1}{R^2}$$

$$= \frac{\rho_V 2\pi}{8\pi \in_0 z^2} \int_{r=0}^{a} \int_{R=z-r}^{z+r} r\left[1 + \frac{z^2 - r^2}{R^2}\right] dR \, dr$$

$$= \frac{\rho_V \pi}{4\pi \in_0 z^2} \left[R - \frac{(z^2 - r^2)}{R}\right]_{z-r}^{z+r} \int_0^a r \, dr$$

$$= \frac{\rho_V \pi}{4\pi \in_0} \int_0^a 4r^2 dr = \frac{1}{4\pi \in_0} \cdot \frac{1}{z^2} \left(\frac{4\pi a^3 \rho_V}{3}\right)$$

$$\vec{E} = \frac{Q}{4\pi \in_0 z^2} \hat{a}_z \quad \text{.....(2.32)}$$

- The electric field at P (r, θ, ϕ) is given by equation (2.32), we get-

$$\vec{E} = \frac{Q}{4\pi \in_0 r^2} \hat{a}_r$$

The above electric field is same as the electric field at the same point due to point charge Q located at the origin or center of spherical charge distribution.

Rack your Brain

If the radius of the Gaussian surface enclosing a charge is halved, how does the electric flux through the Gaussian surface change?

Gray Matter Alert!!!

Gauss's Law helps us in understanding the behavior of electric fields inside the conductors.

ELECTRIC FLUX DENSITY

- The electric flux density vector ($\vec{D}$) in a medium is expressed as the product of the electric field vector and the permittivity of the medium.

  Mathematically, $\vec{D} = \epsilon_0 \vec{E}$ .....(2.33)

  For point charge, the electric field,

  $$\vec{E} = \frac{Q}{4\pi \epsilon_0 R^2} \hat{a}_R$$

  and the electric flux density, $\vec{D} = \frac{Q}{4\pi R^2} \hat{a}_R$

- For line charge, the electric field, $\vec{E} = \frac{\rho_L}{2\pi \epsilon_0 r} \hat{a}_r$

  and the electric flux density, $\vec{D} = \frac{\rho_L}{2\pi r} \hat{a}_r$

- The unit of electric flux density is C/m^2.
- The total flux (ψ) passing through a surface S is defined as the integral of the normal component of $\vec{D}$ through surface,

  $$\psi = \int_S \vec{D} \cdot d\vec{S}$$ .....(2.34)

  where, $\hat{a}_n$ = Unit normal to surface S

  D_n = Component of $\vec{D}$ normal to S.

- For an infinite sheet of charge (ρ_s):

  $$\vec{D} = \frac{\rho_s}{2} \hat{a}_n$$

- For a volume charge distribution (ρ_v):

  $$\vec{D} = \int_V \frac{\rho_v dV}{4\pi R^2} \hat{a}_R$$ .....(2.35)

Example 4:

The electric flux density in the region about a 10 nC/m of uniform line charge which is lying along the z-axis (r = 3m) in the free space will be-

(A) $0.53 \hat{a}_R$ nC/m

(B) $0.61 \hat{a}_R$ C/m

(C) $0.82 \hat{a}_R$ nC/m

(D) $0.53 \hat{a}_R$ C/m

Gray Matter Alert!!!

The Electric Field is dimensionally represented as $[M^1 L^1 I^{-1} T^{-3}]$.

Solution:

Given –

Uniform line charge density

$\rho_L = 10 \text{ nC/m} = 10 \times 10^{-9} \text{ C/m}$

$\rho_L = 10^{-8} \text{ C/m}$

The electric field $\vec{E}$ at this uniform line charge will be–

$$\vec{E} = \frac{\rho_L}{2\pi \epsilon_0 R} \hat{a}_R$$

$$\vec{E} = \frac{10 \times 10^{-9}}{2\pi\left(8.85 \times 10^{-12}\right)R} \hat{a}_R$$

$$\vec{E} = \frac{180}{R} \hat{a}_R$$

At R = 3 m

$\therefore \quad \vec{E} = 60\,\hat{a}_R \text{ V/m}$

The electric flux density $\vec{D}$ associated with the electric field $\vec{E}$ will be –

$\vec{D} = \epsilon_0 \vec{E}$ (for free space)

$\vec{D} = \left(8.85 \times 10^{-12}\right)(60)\hat{a}_R$

$\vec{D} = (0.531)\ \hat{a}_R \text{ nC/m}$

Hence, the correct option is (A).

GAUSS'S LAW

- Statement: "The total electric flux (ψ) passing through any closed surface is equal to the total charge enclosed by that closed surface."
- According to the above statement-

$$\psi = Q_{enc} \qquad \text{..... (2.36)}$$

i.e. $\psi = \oint_S d\psi = \oint_S \vec{D} \cdot d\vec{S}$

- Also, if ρ_V be the volume charge density, then total charge inside the sphere volume will be-

$$Q = \oint_s \vec{D} \cdot d\vec{S} = \int_V \rho_V \, dV \qquad \text{..... (2.37)}$$

Previous Year's Questions

Choose the correct options for the following for Gaussian surface:

(1) A closed surface.

(2) The surface must pass through the point where the electric field is to be calculated.

(3) The surface must have a shape according to the symmetry of the source.

(4) For a system of charges, the Gaussian surface should not pass through any discrete charge but can pass through a continuous charge distribution.

By divergence theorem-

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V \nabla \cdot \vec{D}\, dV \qquad \text{..... (2.38)}$$

- Comparing equation (2.37) and (2.38), the two volume integrals, we have-

$$\int_V \rho_V\, dV = \int_V \nabla \cdot \vec{D}\, dV$$

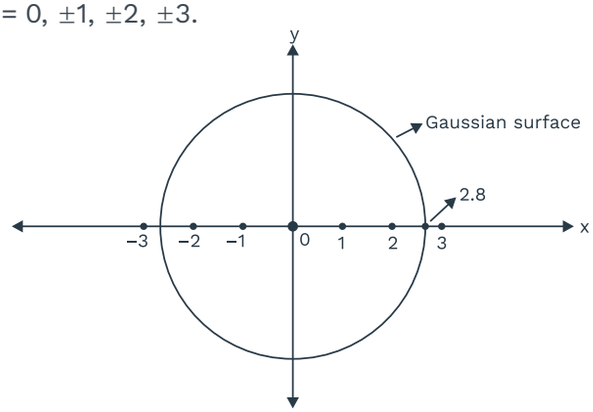
$$\rho_V = \nabla \cdot \vec{D} \qquad \text{.... (2.39)}$$

- Equation(2.39) is referred as the Maxwell's 1st equation in differential form.
  And it states that the volume charge density (ρ_V) is same as the divergence of electric flux density ($\vec{D}$).

Example 5:
The electric flux passing through a sphere of radius 2.8, centered at the origin and the point charge is $Q = 2^{-x^2}\,\mu C$ which is located at x = 0, ±1, ±2, ±3 will be -
(A) 3.125 μC (B) 1.125 μC
(C) 2.125 C (D) 2.125 μC

Solution:

Given, the point charge, $Q = 2^{-x^2}$ and it is located at n = 0, ±1, ±2, ±3.



Thus, according to the Gauss law, the electric flux will be –

$\Psi = Q_{enc}$ and $Q = 2^{-x^2}$ (Given)

Gray Matter Alert!!!

The electric field is dimensionally represented as $[M^1\ L^1\ I^{-1}\ T^{-3}]$.

Previous Year's Questions

Find the flux through a spherical Gaussian surface of radius a = 1 m surrounding a charge of 8.85 pC.
(A) 1 x 10^{-16}N m^2 /C
(B) 1 x 10^{-12} N m^2 /C
(C) 1 x 10^{-8} N m^2 /C
(D) 1 x 10^{-4} N m^2 /C
(E) 1 N m^2 /C

At points x = 0, ±1, ±2

$\therefore \quad \psi = 1 + 2^{-1} + 2^{-1} + 2^{-4} + 2^{-4}$

$\psi = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{16} + \frac{1}{16}$

$\Psi = 2.125\ \mu C$

Hence, the correct option is (D).

APPLICATIONS OF GAUSS LAW

- Gauss law is usually applied where charge distribution has a symmetry.

Electric field due to a point charge

Consider a point charge Q at the origin. To determine $\vec{D}$ or $\vec{E}$ at point A, we choose a spherical surface containing point A, which satisfies the Gaussian symmetry condition.

- Gauss's law is applied to spherical surface S surrounding the point charge Q at the origin and is expressed as-

$$\psi = \oint_S \vec{D} \cdot d\vec{S} = Q_{enclosed} = Q \qquad \text{..... (2.40)}$$

- Since $\vec{D}$ is everywhere and normal to Gaussian surface. Such that $\vec{D} = D_r \hat{a}_r$

Therefore, by Gauss law:

$$Q = \oint_S \vec{D} \cdot d\vec{S} = D_r \oint_S d\vec{S}$$

$$Q = D_r \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin\theta \, d\theta \, d\phi$$

where, $dS = r^2 \sin\theta d\theta d\phi$ for spherical coordinates, D_r is the component of $\vec{D}$

$Q = D_r(4\pi r^2)$

and $\vec{D} = \frac{Q}{4\pi r^2} \hat{a}_r (C/m^2)$ ..... (2.41 (a))

Thus, Electric field will be $\vec{E} = \frac{\vec{D}}{\epsilon_0}$

$\vec{E} = \frac{Q}{4\pi \epsilon_0 r^2} \hat{a}_r (V/m)$ .... (2.41 (b))

Rack your Brain

The electric charge enclosed by Gaussian surface is_____.

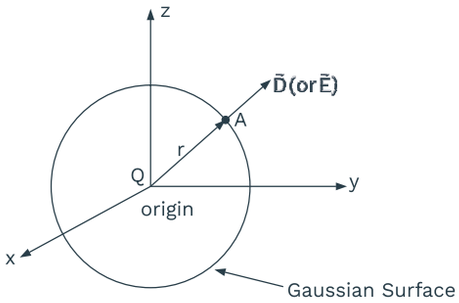


Figure 2.10: Electric field due to point-charge (Q)

Electric field due to infinite line charge distribution

- Considering as infinitely long line charge of density (ρ_L) along the z-axis (from $-\infty$ to ∞).
- Also, assume a right circular cylinder as Gaussian surface, with z-axis as its axis, radius r and length of cylinder be L.

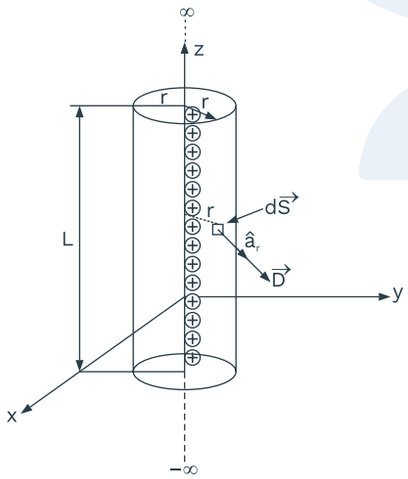


Figure 2.11: Electric field due to infinite line charge.

Rack your Brain

The electric potential V at any point x,y,z (all in meter) in space is given by $V = 4x^2$ volt. The electric field at the point (1m, 0, 2m) is..... V/m

Previous Year's Questions

Three concentric metallic spherical shells of radii R, 2R, 3R, are given with charges Q_1 Q_2, Q_3, respectively. Also the surface charge densities on the outer surfaces of the shells are equal. Then the ratio of the charges given to the shells $Q_1 : Q_2 : Q_3$ is-

(i) 1 : 2 : 3 (ii) 1 : 3 : 5
(iii) 1 : 4 : 9 (iv) 1 : 8 : 18

Previous Year's Questions

Match the following lists-

List–I	**List–II**
A. Gauss's law	(1) $\nabla \times \bar{E} = \frac{-\partial \bar{B}}{\partial t}$
B. Continuity equation	(2) $\nabla \times \bar{H} = J_c + \frac{\partial \bar{D}}{\partial t}$
C. Faraday's law	(3) $\nabla J = \frac{-\partial \rho_V}{\partial t}$
D. Ampere's law	(4) $\nabla \cdot \bar{D} = \rho_V$

Codes:

	A	**B**	**C**	**D**
(a)	4	3	2	1
(b)	4	2	3	1
(c)	3	4	2	1
(d)	4	3	1	2

Rack your Brain

Which of the following is true as per Gauss law- $\int \vec{E} \cdot d\vec{S} = \frac{q_{in}}{\epsilon_0}$

(i) This is valid for symmetrical surface only.

(ii) $\vec{E}$ is the electric field to the charge inside the surface.

(iii) Electric flux on the closed surface due to outside charge is always zero.

(iv) None of the above.

- By symmetry, the $\vec{D}$ (or $\vec{E}$) has the same magnitude at every point on a curved surface and is directed outwards as shown in figure 2.11.
- $\vec{D}$ (or $\vec{E}$) is perpendicular to $d\vec{S}$ everywhere and flux is null.
- For curved surface, (or $\vec{E}$) and $d\vec{S}$ are parallel, then-

$$\vec{D} = D_r \hat{a}_r$$

- Applying Gauss law to the cylindrical surface,

$$\rho_L L = Q = \oint_S \vec{D} \cdot d\vec{S} = D_r \oint_S dS = D_r (2\pi r) L$$

or $\vec{D} = \frac{Q}{2\pi r L} \hat{a}_r = \frac{\rho_L}{2\pi r} \hat{a}_r \ \{as \frac{Q}{L} = \rho_L\}$ ...[2.42(a)]

and Electric field intensity $(\vec{E})$ will be-

$$\vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{\rho_L}{2\pi \epsilon_0 r} \hat{a}_r \text{ V/m} \quad \text{..... [2.42(b)]}$$

Electric field due to infinite sheet

- Consider the infinite sheet of charge with uniform charge density ρ_S (C/m^2), at z = 0 in the x-y plane.
- Also, the Gaussian surface is assumed to be a rectangular box which is cut by the sheet of charge such that it gives, dS = dx·dy.

Rack your Brain

In a region of constant potential-
(i) the electric field is uniform.
(ii) the electric field is zero.
(iii) there can be no charge inside the region.
(iv) both (ii) and (iii) are correct.

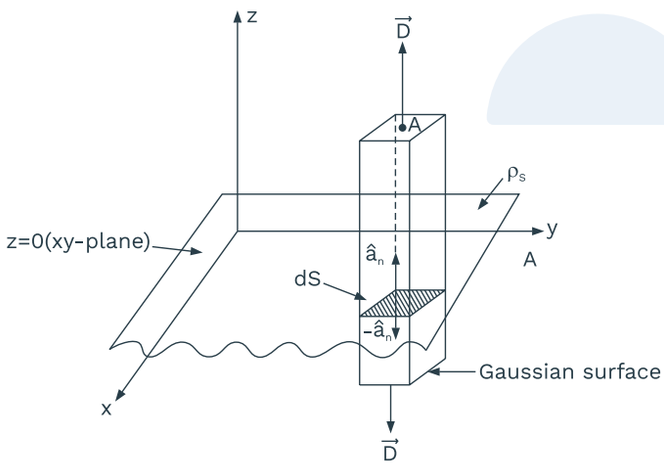


Figure 2.12: Infinite sheet of charge.

- The electric flux density $(\vec{D})$ or electric field $(\vec{E})$ at point A will be evaluated as follows-

The $\vec{D}$ is normal to the sheet, thus-

$\vec{D} = D_z \hat{a}_z$

Now, applying Gauss law,

$$\rho_S \int_S d\vec{S} = Q = \oint_S \vec{D} \cdot d\vec{S} = D_z \left[\int_{top} dS + \int_{bottom} dS \right]$$

- But $\vec{D} \cdot d\vec{S}$ will be zero as $\vec{D}$ has no components in $\hat{a}_x$ and $\hat{a}_y$ directions.
- If both top and bottom area of the box has area A then-

$\rho_S A = D_z [A + A]$

Thus, electric flux density, $\vec{D} = \frac{\rho_S}{2} \hat{a}_z$ ...[2.43(a)]

Electric field, $\vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{\rho_S}{2\epsilon_0} \hat{a}_z$ ...[2.43(b)]

Electric field Intensity due to conducting sphere

- Consider a sphere having radius 'a' and uniform charge density ρ_v (C/m^3).
- For the Gaussian surface, the $\vec{D}$ (or $\vec{E}$) is evaluated separately for the case $r \leq a$ and $r \geq a$.

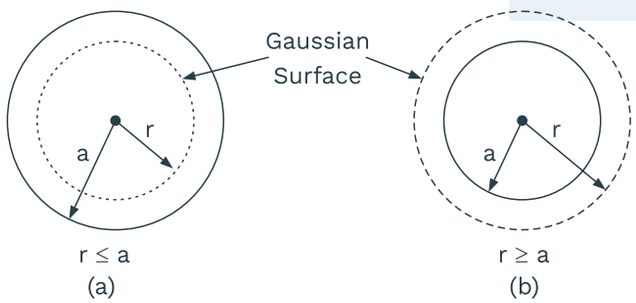


Figure 2.13: Gaussian surface with radius r and sphere with radius a

- For $r \leq a$, the total charge enclosed by spherical surface radius r.

$$Q_{enc} = \int_V \rho_v dV = \rho_v \int_V dV = \rho_v \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$Q_{enc} = \rho_v \frac{4}{3} \pi r^3$$

Gray Matter Alert!!!

A permanent electric dipole is called an electret.

and total flux ψ will be-

$$\psi = \oint_S \vec{D} \cdot d\vec{S} = D_r \oint_S dS = D_r \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin\theta \, d\theta \, d\phi$$

$$\psi = D_r(4\pi r^2)$$

Hence, $\psi = Q_{enclosed}$

Thus, $D_r\left(4\pi r^2\right) = \frac{4\pi r^3}{3}\rho_v$

$$\vec{D} = \frac{r}{3}\rho_v \hat{a}_r \text{ , for } 0 < r \leq a \qquad \text{..... [2.44(a)]}$$

- For $r \geq a$, the charge enclosed will be -

$$Q_{enc} = \int_V \rho_v dV = \rho_v \int_V dV = \rho_v \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{a} r^2 \sin\theta \, d\theta \, d\phi$$

$$Q = \rho_v \frac{4}{3}\pi a^3$$

$$\psi = \int_S \vec{D} \cdot d\vec{S} = D_r(4\pi r^2) = \frac{4}{3}\pi a^3 \rho_v$$

$$\vec{D} = \frac{a^3}{3r^2}\rho_v \hat{a}_r \text{ for } r \geq a \qquad \text{.....[2.44(b)]}$$

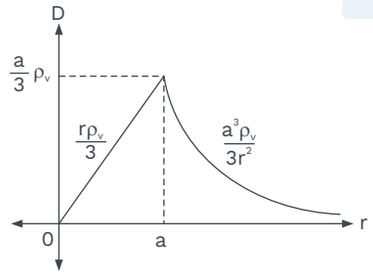


Figure 2.14

- The electric flux density $\vec{D}$ is given by-

$$\vec{D} = \begin{cases} \frac{r}{3}\rho_v \hat{a}_r & ; \quad 0 < r \leq a \\ \frac{a^3}{3r^2}\rho_v \hat{a}_r & ; \quad r \geq a \end{cases}$$

and $|\vec{D}|$ is shown in figure 2.14.

Previous Year's Questions

Consider a positive particle (q=1.0 C) is moving in a uniform electric field (E = 100 V/m) such that it is speeding up. The particle started from rest on an equipotential plane of V = 50 volts. After t = 0.0002 s, the particle is on an equipotential plane of V = 10 Volts. Determine the distance (d) the particle moved.

Rack your Brain

An electric dipole was kept in a uniform electric field then the resultant electric force on it will be-

(a) always null
(b) never zero
(c) depend upon the capacity of dipole
(d) None

Example 6:

A coaxial cable is having inner and outer radii of 2 mm and 4 mm, respectively. The length of the coaxial cable is 60 cm. It is assumed that the space between conductors is filled with air and the total charge on the inner conductor is 20 nC. Determine the electric field density on the conductor inner region for 2 mm < r < 4 mm.

(A) $\frac{5.3}{r}$ nC/m^2 (B) $\frac{-5.3}{r^2}$ nC/m^2

(C) $\frac{6.4}{r^2}$ nC/m^2 (D) $\frac{4.5}{r}$ nC/m^2

Solution:

Given, Length of coaxial cable = 60 cm = 0.6 m

Inner radius (a) = 2 mm = 2×10^{-3} m

Outer radius (b) = 4 mm = 4×10^{-3} m

Total charge on inner conductor.

$Q_{inner} = 20$ nC $= 20 \times 10^{-9}$ C

The surface charge density on the inner cylinder is–

$$\rho_{S(inner)} = \frac{Q_{inner}}{2\pi aL}$$

$$\rho_{S(inner)} = \frac{20 \times 10^{-9}}{2\pi \times \left(2 \times 10^{-3}\right)(0.6)}$$

$$\rho_{S(inner)} = 2.65\ \mu C/m^2$$

The negative charge density of the inner surface of the outer cylinder (i.e. cable) will be –

$$\rho_{S(outer)} = \frac{Q_{outer}}{2\pi bL}$$

$$\rho_{S(outer)} = \frac{-Q_{inner}}{2\pi bL} = \frac{-20 \times 10^{-9}}{2\pi\left(4 \times 10^{-3}\right)(0.6)}$$

$$\rho_{S(outer)} = -1.32\ \mu C/m^2$$

The electric field density on the conductor inner region will be –

Gray Matter Alert!!!

The properties of equipotential surfaces are –

1. Work done in moving a charge over an equipotential surface is zero.
2. The electric field is always perpendicular to an equipotential surface.
3. The spacing between equipotential surfaces enables recognizing the regions of strong and weak fields.
4. Two equipotential surfaces never intersect.

$$\vec{D} = \frac{a\rho_s}{r} = \frac{\left(2\times10^{-3}\right)\left(2.65\times10^{-6}\right)}{r}$$

$$\vec{D} = \frac{5.3\times10^{-9}}{r} \quad \text{(for } 2 < r < 4 \text{ mm)}$$

Note: For $r < 2$ mm & $r > 4$ mm

$$\vec{E} = 0 \quad \& \quad \therefore \ \vec{D} = 0$$

Hence, the correct option is (A).

Rack your Brain

A charged particle (q = 1.6 mC) moves 0.4 m along an equipotential surface of 20V. How much work is done by the field during this motion?

ELECTRIC POTENTIAL

- For determining the electric field intensity $\vec{E}$ due to charge distribution, the coulomb's law has been used earlier.
- Whereas for symmetrical surfaces, the electric field was evaluated using Gauss's law.
- Obtaining $\vec{E}$ from electric scalar potential V is discussed below.
- Electric potential is defined as the amount of work done in moving unit positive charge from one finite distance to another finite distance.
- And the absolute potential is described as the amount of work done in moving unit positive charge from infinity (∞) to a finite distance.
- Potential difference or electric potential is referred to as work done per unit charge.

$$dV = \frac{dW}{dq}$$

$$dW = \vec{F}_{applied} \cdot d\vec{l}$$

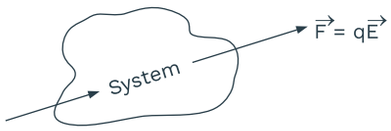


Figure 2.15

- Under equilibrium, $\vec{F}_{applied} + \vec{F} = 0$

or, $\vec{F}_{applied} = -\vec{F} = -q\vec{E}$

Rack your Brain

Does electric field exist without charge density?

$$\vec{F}_{applied} \cdot d\vec{l} = -q\vec{E} \cdot d\vec{l}$$

Work done, $dW = \vec{F}_{applied} \cdot d\vec{l} = -q\vec{E} \cdot d\vec{l}$

$$dV = \frac{dW}{dq} = -\vec{E} \cdot d\vec{l}$$

The electric potential will be-

$$V = -\int \vec{E} \cdot d\vec{l} \quad \text{.... (2.45)}$$

- The potential (V) can be evaluated like as follows.

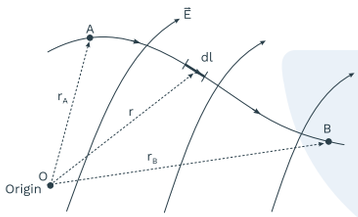


Figure 2.16

$$V_{AB} = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l} = \frac{W}{Q}$$

where,

V_{AB} = Potential difference B with reference to A

V_A = Electric potential at point A

V_B = Electric potential at point B

- The potential at any point r due to a point charge Q located at the origin is-

$$V = \frac{Q}{4\pi \in_0 r}$$

Electric potential due to line charge distribution

Considering a line charge having density ρ_L C/m. And the differential length dl at distance R.

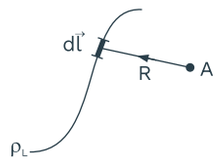


Figure 2.17

Previous Year's Questions

If the current density in a wire is 20.25 A/cm^2 and the electric field in the wire is 10.15 V/cm. Determine the resistivity (ρ), and the conductivity (σ) of the material.

Gray Matter Alert!!!

The electric field potential is continuous across a surface charge and the electric field is discontinuous, but not infinite unless the surface charge consists of a dipole layer.

The differential charge on the line is $dQ = \rho_L \vec{r} dl$ and the voltage at point A due to charge dQ

$$dV = \frac{dQ}{4\pi \in_0 \vec{R}} = \frac{\rho_L \vec{r} dl}{4\pi \in_0 |\vec{r} - \vec{r}'|}$$

where $\vec{R} = |\vec{r} - \vec{r}'|$

The potential due to line charge is-

$$V_L = \frac{1}{4\pi \in_0} \int_L \frac{\rho_L \vec{r} dl}{|\vec{r} - \vec{r}'|}$$

Electric potential due to surface charge distribution

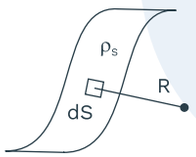


Figure 2.18

Considering a uniform surface-charge having density ρ_S C/m^2. And the differential length dl at distance R.

The differential charge is $dQ = \rho_S \vec{r} dS$

The potential due to surface charge-

$$dV = \frac{dQ}{4\pi \in_0 \vec{R}} = \frac{\rho_s \vec{r} dS}{4\pi \in_0 |\vec{r} - \vec{r}'|}$$

The electric potential due to surface charge will be-

$$V_S = \frac{1}{4\pi \in_0} \int_S \frac{\rho_s \vec{r} dS}{|\vec{r} - \vec{r}'|} \quad .....(2.46)$$

Electric potential due to volume charge distribution

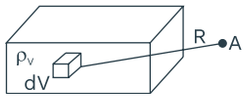


Figure 2.19

- If the volume charge density is ρ_V, then differential charge will be-

Previous Year's Questions

If the potential function V = +3x+2y Volts exists in the free space. Find the stored energy in a 2.3 m^3 volume (spherical) centered at the origin.

$$dQ = \rho_v \vec{r} dV$$

and $$dV = \frac{dQ}{4\pi \epsilon_0 R} = \frac{\rho_v \vec{r} dV}{4\pi \epsilon_0 |\vec{r} - \vec{r}'|}$$

- The total electric potential due to volume charge-

$$V_V = \frac{1}{4\pi \epsilon_0} \int_V \frac{\rho_V \vec{r} dV}{|\vec{r} - \vec{r}'|} \quad \text{..... (2.47)}$$

where $\vec{r}$ is the position vector of V and r′ is the position vector of point considered.

Example 7:

The electric potential field, $V = 3x^2y - 8z$, and point P(4, 2, 3) are considered. Determine the electric field intensity electric flux density and volume charge density.

Solution:

Given, the electric potential-

$$V = 3x^2y - 8z$$

at point P (4, 2, 3)

$$V = 3(4)^2(2) - 8(3)$$

$$V|_{P(4, 2, 3)} \Rightarrow 96 - 24 = 72 \text{ V}$$

The electric field intensity $\vec{E}$ at point P will be –

$$\vec{E} = -\nabla V = -6xy\hat{a}_x - 3x^2\hat{a}_y + 8\hat{a}_z \text{ V/m}$$

at point P(4, 2, 3)

$$\vec{E} = -6(4)(2)\hat{a}_x - 3(4)^2 \hat{a}_y + 8\hat{a}_z$$

$$\vec{E} = -48\hat{a}_x - 48\hat{a}_y + 8\hat{a}_z$$

An electric flux density $\vec{D}$ will be –

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{D} = \epsilon_0 \left(-6xy\hat{a}_x - 3x^2 \hat{a}_y + 8\hat{a}_z\right) \text{C/m}^2$$

$$\vec{D} = -53.1xy\hat{a}_x - 26.55x^2\hat{a}_y + 70.8\hat{a}_z \text{ pC/m}^2$$

At point P(4, 2, 3), $\vec{D}$ will be –

$$\vec{D} = -424.8\hat{a}_x - 424.8\hat{a}_y + 70.8\hat{a}_z \text{ pC/m}^2$$

Previous Year's Questions

The point charges Q_1 = 1.3 nC, Q_2 = -2.7 nC, Q_3 = 4.3 nC, and Q_4 = -4.4 nC are located at (0, 0, 0), (1.1, 0, 0), (0, 0, -2), and (0, 0, 3), respectively. Find the energy in the system.

Also, to calculate volume charge density, we use divergence theorem –

$\therefore \quad \rho_v = \vec{\nabla} \cdot \vec{D}$

$\rho_v = -53.1y \text{ pC/m}^3$

at P(4, 2, 3), $\rho_v = -53.1\,(2)$

$= -106.2 \text{ pC/m}^3$

EQUIPOTENTIAL SURFACES

- The surface or volume over which the electric potential (V) is constant will be referred to as Equipotential surface.
- Consider the volume or surface of a material body or simply a surface or volume of space.

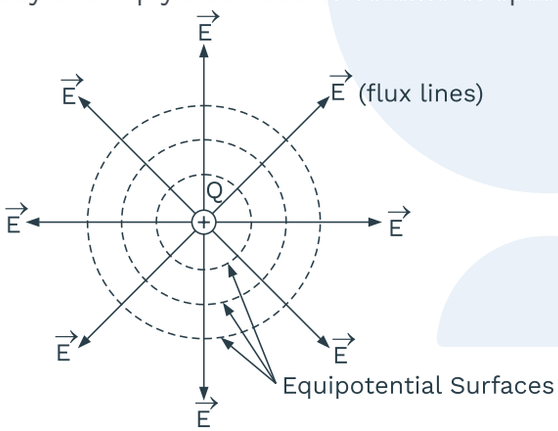


Figure 2.20: Equipotential surface

- There are various points where the electric potential is the same in an electric field.
- This occurs because the potential is a scalar quantity, and it depends on the distance between the point at which the potential is obtained and the location of charge.
- If the surface is illusioned, joining all points where the potentials are equal, then such surface is regarded as Equipotential surface.
- Considering a point charge (Q) located at the origin/center of the sphere, then the potential at a point which is placed at radial distance r from the Q point-charge, is expressed as-

Gray Matter Alert!!!

The potential difference between any two points on an equipotential surface is zero.

Rack your Brain

Equipotential surfaces-

(a) are closer in regions of large electric fields compared to regions of lower electric fields.

(b) will be more crowded near sharp edges of a conductor.

(c) will always be equally spaced.

(d) both (a) and (b) are correct.

No work is done in moving a charge from one point to another along an Equipotential line or surface, then-

$V_A - V_B = 0$

Then, $\int_L \vec{E} \cdot d\vec{l} = 0$ ..... (2.48)

Note: Lines of force/flux $\vec{E}$ are always normal to Equipotential surfaces.

- **Relationship between $\vec{E}$ and V**

  As we know, the electric field is defined as-

  $\vec{E} = E_x\hat{a}_x + E_y\hat{a}_y + E_z\hat{a}_z$ .... (2.49)

  and differential potential will be-

  $dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$ .... (2.50)

- Differential length ($d\vec{l}$) is expressed as-

  $d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$ .... (2.51)

- From equations (2.49) and (2.51), we have-

  $\vec{E} \cdot d\vec{l} = E_x dx + E_y dy + E_z dz$

- Also, we know that-

  $dV = -\vec{E} \cdot d\vec{l}$

  $-\vec{E} \cdot d\vec{l} = -(E_x dx + E_y dy + E_z dz)$ ..... (2.52)

- On comparing (2.52) and (2.50), we get-

  $\frac{\partial V}{\partial x} = -E_x$; $\frac{\partial V}{\partial y} = -E_y$; $\frac{\partial V}{\partial z} = -E_z$

- From equation (2.49)-

  $\vec{E} = -\frac{\partial V}{\partial x}\hat{a}_x - \frac{\partial V}{\partial y}\hat{a}_y - \frac{\partial V}{\partial z}\hat{a}_z$

  $\vec{E} = -\left[\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right]$ .... (2.53)

- Potential gradient is expressed as, V = f(x, y, z)

  $\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$

Previous Year's Questions

If an electric dipole of moment is placed in a uniform electric field $\vec{E}$. Then-
(i) the torque on the dipole is $\vec{T} \times \vec{E}$.
(ii) the potential energy of the system is $\vec{T} \cdot \vec{E}$.
(iii) the resultant force on the dipole is zero.
Choose the correct option from the following-
(a) (i), (ii) and (iii) are correct.
(b) (i) and (iii) are correct and (ii) is wrong.
(c) only (i) is correct.
(d) (i) and (ii) are correct and (iii) is wrong.

- Thus, substituting in equation (2.53), we get-

$$\vec{E} = -\nabla V \quad \text{..... (2.54)}$$

- Equation (2.54) defines that the electric field is a negative gradient of the potential.
- The electric potential difference between two points A and B is independent of the path taken by them.

  Thus, $V_{AB} = -V_{BA}$

  or $V_{AB} + V_{BA} = \oint \vec{E} \cdot d\vec{l} = 0$

  or $\oint_L \vec{E} \cdot d\vec{l} = 0 \quad \text{..... (2.55)}$

- Hence, the line integral of $\vec{E}$ along a closed path is zero. And equation (2.55) is referred to as Maxwell's 2nd equation in integral form.
- Applying stokes theorem in equation (2.55),

  $$\oint_L \vec{E} \cdot d\vec{l} = \int_s (\nabla \times \vec{E}) \cdot d\vec{S} = 0$$

  or, $\nabla \times \vec{E} = 0 \quad \text{..... (2.56)}$

- Any vector field which satisfies the equations (2.55) and (2.56) is said to be conservative/or irrotational.

Rack your Brain

If air is replaced by the dielectric medium of constant K, the maximum force of attraction will -
(a) increases K times.
(b) remains unchanged.
(c) decreases K times.
(d) increases K^{-1} times.

- In another way, those vectors whose line integrals do not depend on the path of integration are regarded as conservative vectors.
  And thus, the electric field is called a conservative field.

Example 8:
If the two charges Q and –2Q are placed at origin and (6, 0) respectively. The equation of the zero equipotential curve in the x-y plane is –
(A) $(x + 2)^2 + y^2 = 16$
(B) $(x - 2)^2 + y^2 = 16$
(C) $x^2 - y^2 - 4x = 12$
(D) $(x + 3)^2 + y^2 = 12$

Solution:
Given, Two-point charges Q and –2Q placed at origin (0, 0) and point (6, 0). The electric potential will be –

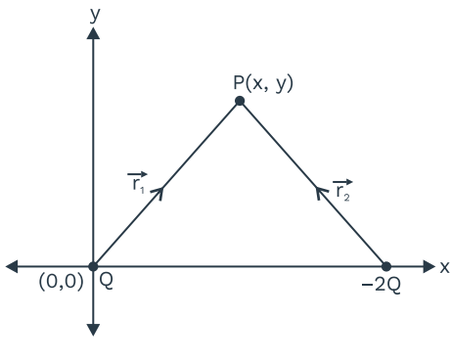
$$V = \frac{kQ}{r_1} + \frac{k(-2Q)}{r_2}$$

$$0 = \frac{1}{r_1} - \frac{2}{r_2}$$

$\therefore \quad 2r_1 = r_2$
As the curve is in x–y plane,

$\therefore \quad r_1 = \sqrt{x^2 + y^2}$ at (0, 0)

and $r_2 = \sqrt{(x-6)^2 + y^2}$ at (6, 0).



Previous Year's Questions

Two identical conducting spheres X and Y, carry an equal charge. They are separated by a distance much larger than their diameters, and the force between them is F. A third identical conducting sphere, Z, is uncharged. Sphere Z is first touched to X, then to Y, and then removed. As a result, the force between X and Y would be equal to-
(a) 3F/8 (b) F/2
(c) 3F/4 (d) F

∴ Substituting the values–

$$2\sqrt{x^2 + y^2} = \sqrt{(x-6)^2 + y^2}$$

$$4x^2 + 4y^2 = x^2 - 12x + 36 + y^2$$

$$3x^2 + 3y^2 + 12x = 36$$

$$x^2 + y^2 + 4x = 12$$

$(x + 2)^2 + y^2 = 16$ is the required equation.

Hence, the correct option is (A).

$\vec{E}$ AND V DUE TO DIPOLE

- If two equal but opposite charges are separated by a finite distance then they create a dipole.
- Dipole is directed from negative to positive charge, unlike the electric field which is usually directed from positive to negative charge.
- The dipole moment is a measurement of the separation of two opposite charges.

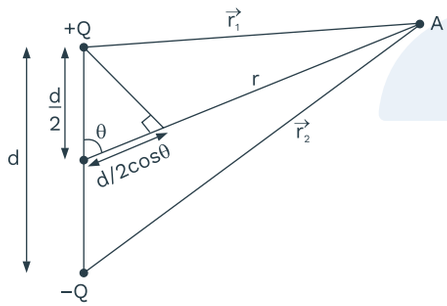


Figure 2.21: An electric dipole

- The potential due to positive charge is given by -

$$V_+ = \frac{Q}{4\pi \epsilon_0 r_1}$$

Similarly, potential due to negative charge will be-

$$V_- = \frac{-Q}{4\pi \epsilon_0 r_2}$$

and $r_1 \simeq r - \frac{d}{2}\cos\theta$

$$r_2 = r + \frac{d}{2}\cos\theta$$

Rack your Brain

Which of the following statement is true?

(a) Electrostatic force is a conservative force.

(b) Potential at a point is the work done per unit charge in bringing a charge from any point to infinity.

(c) Electrostatic force is non-conservative.

(d) Potential is the product of charge and work.

Thus, the overall potential at observation point A is-

$$V = V_+ + V_-$$

$$V = \frac{Q}{4\pi \in_0 r_1} - \frac{Q}{4\pi \in_0 r_2} = \frac{Q}{4\pi \in_0}\left\{\frac{1}{r_1} - \frac{1}{r_2}\right\}$$

$$= \frac{Q}{4\pi \in_0}\left\{\frac{1}{r - \frac{d}{2}\cos\theta} - \frac{1}{r + \frac{d}{2}\cos\theta}\right\}$$

$$V = \frac{Q}{4\pi \in_0}\left\{\frac{d\cos\theta}{r^2 - \left(\frac{d}{2}\cos\theta\right)^2}\right\}$$

Also, $r^2 >>> \left(\frac{d}{2}\cos\theta\right)^2$

Then, $V = \frac{Q}{4\pi \in_0}\left\{\frac{d\cos\theta}{r^2}\right\} = \frac{Qd\cos\theta}{4\pi \in_0 r^2}$

$$V = \frac{Qd\cos\theta}{4\pi \in_0 r^2}$$

- Dipole moment is given by-

$$\vec{p} = Q\vec{d}$$

where, $d\cos\theta = \vec{d}\cdot\hat{a}_z$ and $\vec{d} = d\hat{a}_z$

Thus, $V = \frac{\vec{p}\cdot\hat{a}_r}{4\pi \in_0 r^2}$ ....(2.57)

$\begin{cases}\text{As } \vec{p}\cdot\hat{a}_r = |\vec{p}||\hat{a}_r|\cos\theta \\ \text{or } \vec{p}\cdot\hat{a}_r = Qd\cos\theta\end{cases}$ and $\hat{a}_r = \frac{(r - r')}{|r - r'|}$

- The electric field $(\vec{E})$ will be-

$$\vec{E} = -\nabla V$$

$$= -\left\{\frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta\right\}$$

$$= -\left\{\frac{-2Qd\cos\theta}{4\pi \in_0 r^3}\hat{a}_r - \frac{1}{r}\frac{Qd(\sin\theta)}{4\pi \in_0 r^2}\hat{a}_\theta\right\}$$

Rack your Brain

The electric field for a constant potential region is -
(a) uniform.
(b) zero.
(c) no charge inside the region.
(d) both (b) and (c) are correct.

$$\vec{E} = \frac{-p}{4\pi \in_0 r^3}\left(2\cos\theta\,\hat{a}_r + \sin\theta\,\hat{a}_\theta\right) \quad \ldots\ldots (2.58)$$

where, $p = |\vec{p}| = Qd$

Example 9:

The two dipoles with moments $-6\,\hat{a}_z$ nCm and $10\,\hat{a}_z$ nCm are located at points R(0, 0, –1) and S(0, 0, 4), respectively. The potential at the origin is -

(A) 60 V
(B) 40.5 V
(C) –60 V
(D) – 30.25 V

Solution:

Given, Dipole moments –

$\vec{p}_1 = -6\,\hat{a}_z$ located at R (0, 0, –1)

$\vec{p}_2 = 10\,\hat{a}_z$ located at (0, 0, 4)

$\therefore \quad \vec{r}_1 = (0,0,0) - (0,0,-1)$

$\vec{r}_1 = \hat{a}_z$, $r_1 = |\vec{r}_1| = 1$.

And $\vec{r}_2 = (0,0,0) - (0,0,4)$

$\vec{r}_2 = -4\hat{a}_z$, $r_2 = |\vec{r}_2| = 4$.

Therefore, the potential at the origin will be –

$$V = \sum_{k=1}^{2} \frac{\vec{p}_k \cdot \vec{r}_k}{4\pi \in_0 r_k^3}$$

$$V = \frac{1}{4\pi \in_0}\left[\frac{\vec{p}_1 \cdot \vec{r}_1}{r_1^3} + \frac{\vec{p}_2 \cdot \vec{r}_2}{r_2^3}\right]$$

$$V = \frac{1}{4\pi\left(\frac{10^{-9}}{36\pi}\right)}\left[\frac{-6\hat{a}_z \cdot \hat{a}_z}{(1)^3} + \frac{10\hat{a}_z \cdot (-4\hat{a}_z)}{(4)^3}\right]$$

$$V = \frac{9}{10^{-9}}\left[-\frac{6}{1} - \frac{40}{64}\right]10^{-9}$$

V = –60 Volts

Hence, the correct option is (C).

Previous Year's Questions

At a chargeless boundary between the two dielectric mediums-

(A) There is no electric field.
(B) The tangential electric field component is continuous.
(C) The normal magnetic flux density is discontinuous.
(D) Only the tangential electric field vanishes.

ELECTRIC CURRENT DENSITY ($\vec{J}$)

- The current density ($\vec{J}$) is defined as the amount of current in a unit area.
- Current density is a vector quantity whereas the current is a scalar quantity.
- $\vec{J}$ is usually directed along the direction of flow of current.

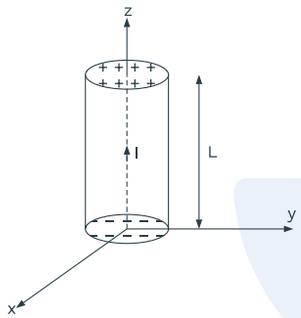


Figure 2.22: Current density

- As described, the current density $\vec{J}$ -

$$\vec{J} = J\hat{a}_z = \frac{dI}{dS}$$

$$dI = \vec{J} \cdot d\vec{S}$$

$$I = \oint \vec{J} \cdot d\vec{S} \qquad \text{..... (2.59)}$$

- And the differential surface is

$$d\vec{S} = d\rho(\rho\, d\phi)\,\hat{a}_z + (\rho\, d\phi\, dz)\,\hat{a}_\rho + dz\, d\rho\, \hat{a}_\phi$$

$$\vec{J} \cdot d\vec{S} = \vec{J} \cdot \hat{a}_z (d\vec{S}) = J\rho\ d\rho\, d\phi$$

From equation (2.59)-

$$I = J \int_0^a \rho\, d\rho \int_{\phi=0}^{2\pi} d\phi$$

$$I = J\left(\frac{\rho^2}{2}\right)_0^a (2\pi) = J\left(\frac{a^2}{2} \times 2\pi\right)$$

$$I = J\pi a^2$$

$$J = \frac{I}{\pi a^2} = \frac{I}{A_C} \qquad \text{..... (2.60)}$$

where A_c = cross sectional area.

Previous Year's Questions

If the point charges -1.9 nC, 3.2 nC, and 5.7 nC are located at (0, 0, 0), (0, 0, 1), and (1, 0, 0), respectively. Evaluate the energy in the system.

- In equation (2.60) the $\vec{J}$ is also referred to as conduction current density according to ohm's law theory.

$$\vec{J} = \sigma\vec{E} \qquad \text{..... (2.61)}$$

Example 10:
The potential function $V(x) = 20\,x^2 + 30\,y^2 + 25\,z^2$ Volts in the free space. The electric magnitude (V/m) and the direction of the electric field at the given point P(–1, 2, –3) will be -

(A) $10\sqrt{385}$; $\dfrac{4\hat{a}_x - 12\hat{a}_y + 15\hat{a}_z}{\sqrt{385}}$

(B) 196 ; $\dfrac{-4\hat{a}_x + 12\hat{a}_y - 15\hat{a}_z}{\sqrt{385}}$

(C) 196 ; $\dfrac{-4\hat{a}_x - 12\hat{a}_y + 15\hat{a}_z}{10\sqrt{385}}$

(D) $\dfrac{\sqrt{385}}{10}$; $\dfrac{4\hat{a}_x - 12\hat{a}_y + 15\hat{a}_z}{10\sqrt{385}}$

Solution:
Given, electric potential function –
$V(x) = 20x^2 + 30y^2 + 25z^2$
Point P (–1, 2, –3)
The electric field $(\vec{E})$ will be –

$$\vec{E} = -\nabla V$$

$$\vec{E} = -\left[\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right]$$

$$\vec{E} = -\left[40x\,\hat{a}_x + 60y\,\hat{a}_y + 50z\,\hat{a}_z\right]$$

Now, at P(–1, 2, –3) the electric field $\vec{E}$ will be –

$$\vec{E} = 40\,\hat{a}_x - 120\,\hat{a}_y + 150\,\hat{a}_z$$

and its magnitude is-

$$|\vec{E}| = \sqrt{(40)^2 + (120)^2 + (150)^2} = 196.21 \text{ V/m}$$

$$|\vec{E}| = 10\sqrt{385} \text{ V/m}$$

Rack your Brain

What happens when two neutral objects are brushed together in a dry environment?

Gray Matter Alert!!!

The electric potential for a system of point charges is equal to the sum of the point charges' individual potentials.

The direction of $\vec{E}$ is–

$$\hat{a}_n = \frac{\vec{E}}{|\vec{E}|} = \frac{10\left(4\hat{a}_x - 12\hat{a}_y + 15\hat{a}_z\right)}{10\sqrt{385}}$$

$$\hat{a}_n = \left(\frac{4\,\hat{a}_x - 12\,\hat{a}_y + 15\,\hat{a}_z}{\sqrt{385}}\right)$$

Hence, the correct option is (A).

ENERGY DENSITY

- The total energy stored in a medium with volume charge density ρ_V, is given by the work done to assemble the charge present in the volume.
- Considering the positioning of three points charges Q_1, Q_2 and Q_3 in an initially empty space.
- No work will be done to move Q, from infinity to P_1 because the space is initially charge-free and there is no electric field i.e. W = 0.
- For transferring Q_2 from infinity to P_2, it is equal to the product of Q_2 and potential V_{21}.

  Similarly for Q_3, it will be $Q_3(V_{31} + V_{32})$.

  Thus, $W_E = W_1 + W_2 + W_3$
  $= 0 + Q_2 V_{21} + Q_3(V_{31} + V_{32})$
- And if the charges were in reverse order, then energy will be-

  $W_E = W_3 + W_2 + W_1$

  $= 0 + Q_2 V_{23} + Q_1(V_{12} + V_{13})$.
- Total energy (W_E) will be-

  $2W_E = Q_1(V_{12} + V_{13}) + Q_2(V_{21} + V_{23}) + Q_3(V_{31} + V_{32})$

  $2W_E = Q_1V_1 + Q_2V_2 + Q_3V_3$

  Thus, $W_E = \frac{1}{2}\left(Q_1V_1 + Q_2V_2 + Q_3V_3\right)$.
- In general, $W_E = \frac{1}{2}\sum_{k=1}^{N} Q_k V_k$ (in Joules)

Gray Matter Alert!!!

The electrostatic field is a vector quantity & often expressed as the gradient of the electrostatic potential(V), a scalar quantity.
It is equal to the potential energy of any charged particle at any location (J) divided by the charge of that particle (C).

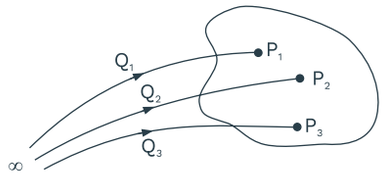


Figure 2.23: Assembling of charges

- Energy in different charge distributions-

$$W_E = \frac{1}{2}\int_L \rho_L V\, dl \quad \text{(Line charge)}$$

$$W_E = \frac{1}{2}\int_S \rho_S V\, dS \quad \text{(Surface charge)}$$

$$W_E = \frac{1}{2}\int_V \rho_v V\, dV \quad \text{(Volume charge)}$$

- Since, $\rho_v = \nabla \cdot \vec{D}$, $E = -\nabla V$ and $\vec{D} = \epsilon_0 \vec{E}$

$$W_E = \frac{1}{2}\int_V (\nabla \cdot \vec{D})\, V\, dV$$

$$= \frac{1}{2}\int_V (\vec{E} \cdot \vec{D})\, dV = \frac{1}{2}\int_V \epsilon_0 E^2\, dV$$

- Electrostatic energy density w_E (in J/m^3) is expressed as-

$$w_E = \frac{dW_E}{dV} = \frac{1}{2}(\vec{D} \cdot \vec{E}) = \frac{1}{2}\epsilon_0 E^2 = \frac{D^2}{2\epsilon_0} \quad \text{.... (2.62)}$$

or $W_E = \int_V w_E dV$

Example 11:
A charge distribution with spherical symmetry has a volume charge density described as–

$$\rho_v = \begin{cases} \rho_0, & 0 \le r \le R \\ 0, & r > R \end{cases}$$

The energy stored in the region $r \le R$ will be -

(A) $\frac{2\pi R^5 \rho_0^{\,2}}{45\,\epsilon_0}$ J

(B) $\frac{4\pi R^2 \rho_0^{\,2}}{18\,\epsilon_0}$ J

(C) $\frac{2\pi \rho_0^{\,2} R^2}{9\,\epsilon_0}$ J

(D) $\frac{2\pi \rho_0^{\,2} R^2}{45\,\epsilon_0}$ J

Gray Matter Alert!!!

The energy density is dimensionally represented as $[M^1\ L^{-1}\ T^{-2}]$.

Rack your Brain

A test charge is moved from lower potential point to a higher potential point. The potential energy of test charge will-
(a) remain the same
(b) increase
(c) decrease
(d) become zero

Solution:

Given that, Volume charge density–

$$\rho_v = \begin{cases} \rho_0, & 0 \le r \le R \\ 0, & r > R \end{cases}$$

The energy stored in $r \le R$ will be–

$$W = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}\,dV = \frac{1}{2}\epsilon_0 \int_V E^2 dV$$

As $\vec{E} = \frac{\rho_0 r}{3\epsilon_0}\hat{a}_r$ for $r \le R$

$$\therefore \quad W = \frac{1}{2}\epsilon_0 \int_V E^2 dV$$

$$W = \frac{1}{2}\epsilon_0 \int_{r=0}^{R}\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi}\left(\frac{\rho_0 r}{3\epsilon_0}\right)^2 r^2 \sin\theta\, d\phi\, d\theta\, dr$$

$$W = \frac{1}{2}\epsilon_0 \left(\frac{\rho_0^{\,2}}{9(\epsilon_0)^2}\right)\int_{r=0}^{R}\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} r^4 \sin\theta\, d\phi\, d\theta\, dr$$

$$W = \frac{\rho_0^{\,2}}{18\epsilon_0}\times\int_{r=0}^{R} r^4 dr \int_{\theta=0}^{\pi}\sin\theta\, d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$W = \frac{\rho_0^{\,2}}{18\epsilon_0}(4\pi)\left[\frac{r^5}{5}\right]_0^R$$

$$W = \frac{2\pi\rho_0^{\,2}R^5}{45\epsilon_0}\text{ Joules}$$

Hence, the correct option is (A).

CHAPTER SUMMARY



- Coulomb's law is given by-

$$\vec{F} = \frac{1}{4\pi \epsilon_0} \frac{Q_1 Q_2}{R^2} \hat{a}_R$$

- Electric field intensity at a point due to charge Q at distance 'R' is given by-

$$\vec{E} = \frac{1}{4\pi \epsilon_0} \frac{Q}{R^2} \hat{a}_R$$

- Electric field due to an infinite line charge is given by-

$$\vec{E} = \frac{\rho_L}{2\pi \epsilon_0 R} \hat{a}_R$$

- Electric field due to an infinite sheet of surface charge is given by-

$$\vec{E} = \frac{\rho_S}{2 \epsilon_0} \hat{a}_n$$

- Gauss's Divergence theorem is given by-

$$\nabla \cdot \vec{D} = \rho_V$$

- Scalar potential is related to the electric field as-

$$V = -\int \vec{E} \cdot d\vec{l}$$

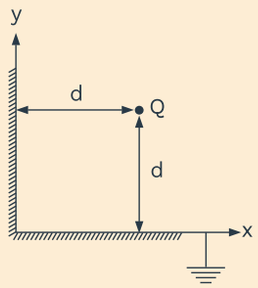
or $\vec{E} = -\nabla V$

- Energy density in the electrostatic field is given as-

$$W_E = \frac{1}{2} \int_V \epsilon_0 E^2 dV$$

PRACTICE QUESTIONS

Q1 Two semi-infinite conducting sheets are placed at right angles to each other as shown in the figure. A point charge of +Q is placed at d distance from both sheets. The net force on the charge is $\frac{Q^2}{4\pi\varepsilon_0}\frac{K}{d^2}$, where K is given by-



(A) 0

(B) $-\frac{1}{4}\hat{i}-\frac{1}{4}\hat{j}$

(C) $-\frac{1}{8}\hat{i}-\frac{1}{8}\hat{j}$

(D) $\frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{i}+\frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{j}$

Q2 The two-point charges are located at $z=\pm\frac{d}{2}$ having equal magnitude but opposite signs. The electric field intensity at the origin is-

(A) $\frac{+2Q}{4\pi\in_0 d^3}\hat{a}_z$ V/m

(B) $\frac{-2Q}{\pi\in_0 d^2}\hat{a}_z$ V/m

(C) $\frac{Q}{2\pi\in_0 d^3}\hat{a}_z$ V/m

(D) $\frac{-Q}{2\pi\in_0 d^2}\hat{a}_z$ V/m

Q3 The electric flux density is described as $\vec{D}=5(r-3)^3\hat{a}_r$ C/m² within the spherical shall (3 < r < 4 m). The volume charge density at r = 4 is -

(A) 17.5 μ C/m³

(B) 15 C/m³

(C) 17.5 C/m³

(D) 12 mC/m³

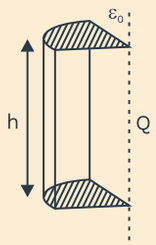
Q4 The charge density of a circular disk enclosed by surface P having diameter 8m is 22 sin ϕ $\mu C/m^2$. The net flux crosses the surface P is -

(A) 22 μC
(B) 176 μC
(C) 0 C
(D) 2.75 μC

Q5 Consider a (+Q) point charge located at origin and the electric field due to this point charge at far point R is 100 mV/m. If the given point charge is enclosed by the perfectly conducting hollow metallic sphere centered at origin, then the electric field at point R will have one of the following features -

(A) Its magnitude will be changed but the direction remains unchanged.
(B) Its magnitude remains unchanged but its direction will be reversed.
(C) It will be zero.
(D) None of the above.

Q6 Consider a quarter-cylinder of height h in which an infinitely long line charge is placed along the cylindrical axis with Q charge density. The electric flux through the given cylinder will be –



(A) $\frac{hQ}{2\varepsilon_0}$
(B) $\frac{h\varepsilon_0}{2Q}$
(C) $\frac{4h}{Q\varepsilon_0}$
(D) $\frac{hQ}{4\varepsilon_0}$

Q7 In a free space region, the electric field is given as –

$\vec{E} = -\left(2y^3 - 3yz^2\right)\hat{a}_x - \left(6xy^2 - 3xz^2\right)\hat{a}_y + 6xy\,\hat{a}_z V/m$. The electric potential for the same region will be -

(A) $2x^3 + xy^2z$
(B) $2xy^3 - 3xyz$
(C) $x^4y^2 + x^2z$
(D) $3xy^3 - 2xyz$

Q8 If 3 identical 4 pC point charges are located at the corners of an equilateral triangle with 0.5 mm side in the free space, then the work done to move a charge to the equidistant from the other two charges and the line joining them must be -

(A) 400 J (B) 576 pJ (C) 500 J (D) 325.15 pJ

Q9 Consider a dipole with dipole moment $\vec{p} = 3\hat{a}_x - 5\hat{a}_y + 10\hat{a}_z$ ncm which is located at Q(1, 2, –4) in the free space. The potential at the point P(2, 3, 4) will be-

(A) 1 V (B) 1.5 V (C) 1.31 V (D) 2 V

Q10 Consider an electric field intensity in the free space that exhibits a Gaussian function given as –

$$\vec{E} = \frac{\rho_0 a}{\epsilon_0} e^{-\frac{r^2}{a^2}} \hat{a}_r$$

where ρ_0 and a are constants and field exists everywhere.
The net energy (J) stored in the field will be -

(A) $\frac{2\pi a \rho_0^{\,2}}{3\epsilon_0}$

(B) $\frac{a\pi\rho_0^{\,2}}{2\epsilon_0}\left(\frac{\pi}{2}\right)$

(C) $\frac{a^5\pi\rho_0^{\,2}}{4\epsilon_0}\left(\sqrt{\frac{\pi}{2}}\right)$

(D) $\frac{a^2\pi\rho_0^{\,5}}{6\epsilon_0}\left(\sqrt{\frac{\pi}{2}}\right)$

ANSWER KEY:

Q1	(D)	**Q2**	(B)	**Q3**	(C)	**Q4**	(C)	**Q5**	(C)
Q6	(D)	**Q7**	(B)	**Q8**	(B)	**Q9**	(C)	**Q10**	(B)